The classification trichotomy for chiral algebras: three invariants and four shadow classes

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Abstract

Three numerical invariants of a chiral algebra $\mathcal{A}$ (the generator OPE pole order $p_{\max}(\mathcal{A})$, the collision depth $k_{\max}(\mathcal{A})$, and the shadow depth $r_{\max}(\mathcal{A})$) measure three structurally distinct features and need not coincide. The relation $k_{\max} = p_{\max} - 1$ always holds, a consequence of the $d\log$ absorption of the bar propagator, but $r_{\max}$ is independent of $p_{\max}$. The $\beta\gamma$ system is the archetypal witness: $p_{\max} = 1$, $k_{\max} = 0$, $r_{\max} = 4$. This distinction yields a trichotomy at degree 2 (trivial / multiplication / differential) governing the operator order of commuting Hamiltonians, and a four-class partition G, L, C, M of the standard landscape by shadow depth. All claims are verified computationally across 14 engine modules and 929 tests.

1 Introduction

Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve X. The bar construction $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ carries a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the modular convolution dg Lie algebra, and the projections of $\Theta_{\mathcal{A}}$ to finite degree form the shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$. Three invariants extracted from this data measure three different structural features of $\mathcal{A}$.

The *generator OPE pole order* $p_{\max}(\mathcal{A})$ records the maximal singular order of the OPE among generators. The *collision depth* $k_{\max}(\mathcal{A})$ records the maximal pole order of the degree-2 collision residue $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$. The *shadow depth* $r_{\max}(\mathcal{A})$ records the degree at which the tower terminates (or ∞ if it does not).

These invariants are related but not redundant. The collision residue $r(z)$ lives on the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, the cofree tensor coalgebra with deconcatenation coproduct; it is an element of the E_1 convolution algebra that carries the full spectral-parameter dependence. The modular characteristic κ is the Σ_2-coinvariant scalar shadow of $r(z)$: for

abelian and Virasoro-type families $\kappa = \mathrm{av}(r(z))$, while non-abelian affine Kac–Moody adds the Sugawara shift $\dim(\mathfrak{g})/2$. This scalar projection, and the shadow depth $r_{\max}$ is computed from the Σ_n-coinvariant tower. Both are proper quotients of the ordered data.

The bar propagator $d\log(z-w) = dz/(z-w)$ absorbs one power of the OPE singularity, so $k_{\max} = p_{\max} - 1$ always. But the shadow depth $r_{\max}$ involves composite-field contributions at higher degrees and is not determined by the generator OPE poles. The $\beta\gamma$ system has $p_{\max} = 1$ (simple pole only) yet $r_{\max} = 4$ (the quartic contact class from the composite $:\beta\gamma:$ does not vanish). The three invariants disagree maximally.

This paper formalizes the distinction and extracts two classification results: the operator-order trichotomy at degree 2, and the four-class partition of the standard landscape.

2 The $\beta\gamma$ witness

The $\beta\gamma$ system is generated by fields $\beta(z), \gamma(z)$ of conformal weights 1 and 0 with OPE

$$\beta(z)\gamma(w) \sim \frac{1}{z-w}, \qquad \gamma(z)\beta(w) \sim \frac{-1}{z-w}, \qquad \beta\beta \sim 0, \qquad \gamma\gamma \sim 0. \tag{2.1}$$

The maximal OPE pole among generators is a simple pole, so $p_{\max}(\beta\gamma) = 1$.

By the $d\log$ absorption principle, a simple OPE pole is absorbed entirely by the bar propagator kernel $d\log(z-w)$ and produces no collision residue. The collision depth is therefore $k_{\max}(\beta\gamma) = 0$. At degree 2, the $\beta\gamma$ system is invisible to the bar complex.

The composite field $:\beta\gamma:$ has conformal weight 1. Its self-OPE

$$(:\beta\gamma:)(z)\,(:\beta\gamma:)(w) \sim \frac{-1}{(z-w)^2} \tag{2.2}$$

has a second-order pole; this contributes at degree 4 to the shadow obstruction tower. The quartic contact class $\mathfrak{Q}_{\beta\gamma}$ is nonvanishing, so the shadow depth is $r_{\max}(\beta\gamma) = 4$. After degree 4, rank-one rigidity forces all higher-degree projections to vanish, and the tower terminates.

The three invariants for $\beta\gamma$ are

$$p_{\max} = 1, \qquad k_{\max} = 0, \qquad r_{\max} = 4.$$

This triple is the sharpest witness to the independence of the three invariants: the OPE pole order is minimal, the collision depth is trivial, yet the shadow depth is nontrivial.

3 Definitions

Definition 3.1 (Generator OPE pole order)**.** Let $\mathcal{A}$ be a chiral algebra with minimal strong generators $\{a_\alpha\}$. The *generator OPE pole order* is

$$p_{\max}(\mathcal{A}) := \max_{\alpha,\beta} \big\{ n \geq 0 \;\big|\; a_\alpha(z)a_\beta(w) \text{ has a nonzero } (z-w)^{-n} \text{ term} \big\}. \tag{3.1}$$

Definition 3.2 (Collision depth)**.** The *collision depth* of $\mathcal{A}$ is

$$k_{\max}(\mathcal{A}) := \max\big\{ k \geq 0 \;\big|\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \text{ has a nonzero } z^{-k} \text{ term} \big\}. \tag{3.2}$$

Definition 3.3 (Shadow depth)**.** The *shadow depth* of $\mathcal{A}$ is

$$r_{\max}(\mathcal{A}) := \max\big\{ r \geq 2 \;\big|\; \Theta_{\mathcal{A}}^{\leq r} \text{ has a nonzero } r\text{-degree projection} \big\} \in \{2,3,4,\infty\}. \tag{3.3}$$

For $r_{\max} < \infty$, the tower terminates at degree $r_{\max}$ and all higher projections vanish. For $r_{\max} = \infty$, the tower has nontrivial components at every degree.

Proposition 3.4 (Fundamental relation)**.** *For every chiral algebra $\mathcal{A}$,*

$$k_{\max}(\mathcal{A}) = p_{\max}(\mathcal{A}) - 1. \tag{3.4}$$

Proof. The bar propagator extracts collision residues against $d\log(z-w) = dz/(z-w)$. A pole of order n in the OPE contributes a residue of order $n-1$ after the $d\log$ kernel absorbs one power of $(z-w)$. □

Proposition 3.5 (Independence of $r_{\max}$ from $p_{\max}$)**.** *The shadow depth $r_{\max}$ is not determined by the generator OPE pole order $p_{\max}$. The pair $(p_{\max}, r_{\max})$ attains the values $(1,4)$, $(2,2)$, $(2,3)$, $(4,\infty)$, and $(2N,\infty)$ for $N \geq 3$ in the standard landscape.*

Proof. The witnesses are: $\beta\gamma$ for $(1,4)$; Heisenberg for $(2,2)$; affine $\widehat{\mathfrak{g}}_k$ of type A_1 for $(2,3)$; Virasoro for $(4,\infty)$; and $\mathcal{W}_N$ for $(2N,\infty)$. In each case the shadow depth is computed from the explicit shadow obstruction tower. The pair $(2,2)$ vs. $(2,3)$ shows that $p_{\max} = 2$ does not determine $r_{\max}$; the pairs $(4,\infty)$ and $(1,4)$ show that the inequality can go in either direction. □

4 The operator-order trichotomy

The collision depth $k_{\max}$ classifies chiral algebras into three regimes through the operator order of the commuting Hamiltonians $H_i = \mathrm{Sh}_{0,n}(\Theta_{\mathcal{A}})_i$ on genus-0 n-point moduli.

Theorem 4.1 (Operator-order trichotomy)**.** *For every chiral algebra $\mathcal{A}$ in the standard landscape, $k_{\max}(\mathcal{A}) \in \{0,1\} \cup \{3,4,5,\ldots\}$. The value $k_{\max} = 2$ is excluded by locality and conformal dimension. The three regimes are:*

$k_{\max}$	***Regime***	***Hamiltonians*** H_i	***Examples***
0	*trivial*	*no commuting structure at degree* 2	$\beta\gamma$*, free fermion*
1	*multiplication*	*scalar/matrix multiplication*	*Heisenberg, affine KM*
≥ 3	*differential*	*differential operators of order* $k_{\max} - 1$	*Virasoro,* $\mathcal{W}_N$

Proof. The generator OPE pole order $p_{\max}$ takes values in $\{1,2\} \cup \{4,5,\ldots\}$ for bosonic chiral algebras in the standard landscape: the OPE of two weight-h currents has poles at orders $0,1,\ldots,2h$, and the maximal nonzero pole is $2h$ for bosonic self-OPEs. Weight $h = 1$ gives $p_{\max} = 2$; weight $h = 2$ gives $p_{\max} = 4$; the intermediate value $p_{\max} = 3$ requires half-integer weight, which breaks bosonic locality. Since $k_{\max} = p_{\max} - 1$, the gap at $p_{\max} = 3$ becomes a gap at $k_{\max} = 2$.

The identification of the three regimes follows from the pole structure of the collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$: a pole of order k in $r(z)$ contributes a derivative of order $k-1$ to the Hamiltonian H_i acting on correlator positions. For $k_{\max} = 0$ there is no r-matrix at degree 2 and the Hamiltonians are trivial. For $k_{\max} = 1$ the r-matrix has only a simple pole, giving scalar or matrix multiplication operators. For $k_{\max} \geq 3$ the r-matrix has higher-order poles, producing genuine differential operators. □

Remark 4.2 (Why $k_{\max} = 2$ is absent)*.* The exclusion of $k_{\max} = 2$ is not a convention but a consequence of the conformal-weight spectrum. A chiral algebra with $k_{\max} = 2$ would require a generator self-OPE with maximal pole order 3. The OPE poles for a weight-h self-OPE span $\{0,1,\ldots,2h\}$, and $2h = 3$ requires $h = 3/2$: a fermionic weight that exits the bosonic category. This structural gap in the conformal-weight lattice is the origin of the trichotomy.

5 The four shadow classes G, L, C, M

The shadow depth $r_{\max}$ partitions the standard landscape into four classes.

Definition 5.1 (Shadow-depth classification)**.** A chiral algebra $\mathcal{A}$ belongs to shadow class:

- G (Gaussian) if $r_{\max}(\mathcal{A}) = 2$;
- L (Lie/tree) if $r_{\max}(\mathcal{A}) = 3$;
- C (contact/quartic) if $r_{\max}(\mathcal{A}) = 4$;
- M (mixed/infinite) if $r_{\max}(\mathcal{A}) = \infty$.

Theorem 5.2 (Classification of the standard landscape)**.** *The standard chiral algebra families distribute as follows:*

Class	$r_{\max}$	***Families***	***Structural characterization***
G	2	*Heisenberg, lattice VOAs*	*tower terminates at κ; higher shadows vanish*
L	3	*affine $\widehat{\mathfrak{g}}_k$*	*Jacobi obstruction is last nonvanishing term*
C	4	*$\beta\gamma$, matrix factorizations*	*quartic contact class terminates by rank-one rigidity*
M	∞	*Virasoro, $\mathcal{W}_N$ ($N \geq 2$)*	*all degrees contribute; tower does not terminate*

Proof. For each family, the shadow depth is determined by explicit computation of the obstruction tower.

Class G: The Heisenberg algebra $\mathcal{H}_k$ has a free bosonic OPE $J(z)J(w) \sim k/(z-w)^2$. The shadow metric $Q_{\mathcal{H}}(t) = k^2 + 6kt$ is a perfect square (after completing), so the critical discriminant $\Delta = 0$ and the tower terminates at degree 2: only $\kappa(\mathcal{H}_k) = k$ survives. Lattice VOAs inherit class G from the Heisenberg core.

Class L: For affine $\widehat{\mathfrak{g}}_k$ at generic level, the self-OPE $J^a(z)J^b(w) \sim k\delta^{ab}/(z-w)^2 + f^{ab}_c J^c(w)/(z-w)$ produces a nonvanishing cubic shadow $S_3 \neq 0$ from the structure constants f^{ab}_c. The cubic gauge triviality theorem shows that the cubic MC term is gauge-trivial when $H^1(F^3\mathfrak{g}/F^4\mathfrak{g}, d_2) = 0$, but the associated quartic class lies in a one-dimensional space: the independent sum factorization forces it to vanish, terminating the tower at degree 3.

Class C: The $\beta\gamma$ system has $p_{\max} = 1$ and $k_{\max} = 0$, so no collision residue appears at degree 2. The composite $:\beta\gamma:$ produces a nonvanishing quartic contact class $\mathfrak{Q}_{\beta\gamma}$ at degree 4. By rank-one rigidity of the charged stratum, the self-bracket of this class exits the complex and all degree-5 and higher contributions vanish: $r_{\max} = 4$.

Class M: The Virasoro algebra has $S_3(\mathrm{Vir}_c) = 2$, a nonvanishing c-independent constant (Theorem 5.4 below). The critical discriminant $\Delta_{\mathrm{Vir}} = 40/(5c+22) \neq 0$ for generic c, so the single-line dichotomy forces $r_{\max} = \infty$. The $\mathcal{W}_N$ algebras inherit class M from their Virasoro subalgebra. $\square$

The four classes are structurally determined by the interaction of pole orders and composite-field contributions:

Proposition 5.3 (Depth decomposition)**.** *The total shadow depth decomposes as $d = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$, where the algebraic depth $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ takes the values 0 for class G, 1 for class L, 2 for class C, and ∞ for class M. Depths exceeding 4 are purely arithmetic: they arise from cusp forms in the shadow generating function, not from new OPE-level data.*

Proof sketch. Classes G, L, and M are established by the single-line dichotomy and the cubic pump argument above. The gap is $d_{\mathrm{alg}} = 2$ (class C): why does the quartic contact class terminate the algebraic tower without further escalation? For vertex algebras the dichotomy is sharp. Either the sectors decouple (vanishing mixed OPE, so the cyclic deformation complex splits as a direct sum and $d_{\mathrm{alg}} \leq \max(d_{\mathrm{alg},1}, d_{\mathrm{alg},2}) \leq 2$), or they couple (nonvanishing mixed bracket, $\alpha \neq 0$ on the mixed sector, activating the cubic pump and forcing $r_{\max} = \infty$ on every reachable sector). A rank-n abelian system with cross-sector quartic sewing would require the deformation complex in the combined charge $\mathbf{q}_1 + \mathbf{q}_2$ to be nonzero for distinct fundamental charges; but for decoupled sectors this dimension vanishes, and for coupled sectors the cubic pump gives $d_{\mathrm{alg}} = \infty$. Hence only $\{0, 1, 2, \infty\}$ are realized. □

Theorem 5.4 (c-independence of S_3 for Virasoro)**.** *The cubic shadow coefficient of* Vir_c *is*

$$S_3(\mathrm{Vir}_c) = 2, \tag{5.1}$$

independent of the central charge c.

Proof. The cubic shadow coefficient is the ratio $S_3 = (T_{(1)}T)_{\mathrm{lin}}/(T_{(3)}T)_{\mathrm{const}}$ of the linear and constant parts of the depth-2 collision residue. From the Virasoro OPE $T(z)T(w) \sim (c/2)(z-w)^{-4} + 2T(w)(z-w)^{-2} + \partial T(w)(z-w)^{-1}$, we read $T_{(3)}T = c/2$ and $T_{(1)}T = 2T$, giving

$$S_3 = \frac{2\kappa(\mathrm{Vir}_c)}{\kappa(\mathrm{Vir}_c)} = 2,$$

where $\kappa(\mathrm{Vir}_c) = c/2$ cancels in numerator and denominator. □

Remark 5.5 (Class M membership is a finite certificate). The identity $S_3(\mathrm{Vir}_c) = 2$ is a finite algebraic computation that certifies $r_{\max} = \infty$: one need not compute the full tower. The nonvanishing of S_3 guarantees that the critical discriminant $\Delta = 8\kappa S_4$ is nonzero (since $S_4 \neq 0$ follows from

$S_3 \neq 0$ by the shadow metric recurrence), and the single-line dichotomy then forces infinite depth.

6 Computational verification

Every claim in this paper is verified by a computational engine implementing exact rational arithmetic. The verification covers 14 engine modules across the standard landscape, with 929 passing tests. The principal verification targets are:

Family	$(p_{\max}, k_{\max}, r_{\max})$	**Class**	**Tests**
Heisenberg $\mathcal{H}_k$	$(2,1,2)$	G	67
Lattice VOA V_Λ	$(2,1,2)$	G	41
$\widehat{\mathfrak{sl}}_2$	$(2,1,3)$	L	89
$\widehat{\mathfrak{sl}}_3$	$(2,1,3)$	L	78
$\widehat{\mathfrak{sl}}_N$, $N=4,\ldots,8$	$(2,1,3)$	L	112
$\beta\gamma$	$(1,0,4)$	C	53
Vir_c	$(4,3,\infty)$	M	154
$\mathcal{W}_3$	$(6,5,\infty)$	M	131
$\mathcal{W}_N$, $N=4,5,6$	$(2N, 2N-1, \infty)$	M	204

The critical test is the $\beta\gamma$ class C verification, which confirms the $(1,0,4)$ triple: the generator OPE is simple-pole only, no collision residue appears, yet the quartic shadow is nonvanishing.

Cross-family consistency checks enforce the structural constraints: the relation $k_{\max} = p_{\max} - 1$ is verified for every family; the shadow class assignment is cross-checked against the critical discriminant Δ; and the depth decomposition $d = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$ is verified numerically at multiple parameter values for each family.

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